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Physical Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Thermodynamics-01

English Medium

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THERMODYNAMICS-01

1. INTRODUCTION

Thermodynamics is concerned with energy and its transformation in various forms in different physical and chemical processes.

Thermodynamics ≡ Thermo + dynamics
Dynamics ≡ Study of change due to a driving force
Thermo ≡ Thermal which is related to temperature or energy.

The laws of thermodynamics deal with energy changes of macroscopic systems involving a large number of molecules rather than microscopic systems containing a few molecules. Thermodynamics is not concerned about how and at what rate these energy transformations are carried out, but is based on initial and final states of a system undergoing the change. Laws of thermodynamics apply only when a system is in equilibrium or moves from one equilibrium state to another equilibrium state. Macroscopic properties like pressure and temperature do not change with time for a system in equilibrium state.

2. SOME BASIC DEFINITITIONS (Thermodynamic terms)

2.1 SYSTEM :

The macroscopic part of the universe under study is called a **system**. Rest of the universe outside the system is called **surroundings**. The actual or imaginary surface that separates the system from the surroundings is called the **boundary**.

However, the entire universe other than the system is not affected by the changes taking place in the system. Therefore, for all practical purposes, the surroundings are that portion of the remaining universe which can interact with the system. Usually, the region of space in the neighbourhood of the system constitutes its surroundings.

2.1.1 Types of boundary :

(i) Real or imaginary
(ii) Rigid (fixed) or movable (flexible)
(iii) Permeable (allow mass transfer) or impermeable (does not allow mass transfer).
(iv) Diathermal (allow heat transfer) or Adiabatic (does not allow heat transfer).

2.1.2 Types of systems :

(i) Isolated system : A system is said to be isolated if it cannot exchange both matter and energy with the surroundings. Example : coffee in a thermos flask.

(ii) Closed system : A system is said to be closed if it can exchange energy but not matter. Example : Coffee in a closed stainless steel flask.

(iii) Open system : A system is said to be open if it can exchange matter and energy. Example : A thermos flask or a steel flask if not closed.

Note : A perfectly isolated system is only a theoretical system.

2.2 State of system : A system is called in a particular state where all the macroscopic properties of the system have definite value.

2.3 Properties of system : The state of a system is defined by a particular set of its measurable quantities called properties. They can be categorised into extensive and intensive properties.

Intensive property is one whose value is independent of the size (or mass) of the system. An extensive property is one whose value depends on the size (or mass) of the system.

* Extensive properties are additive but intensive propreties are non additive.
* Ratio of two extensive property gives an intensive property.

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Extensive Properties	Intensive Properties
Volume	Molar volume
Number of moles	Density
Mass	Refractive index
Free Energy (G)	Surface tension
Entropy (S)	Viscosity
Enthalpy (H)	Free energy per mole
Internal energy (E & U)	Specific heat
Heat capacity	Pressure
	Temperature
	Boilling point, freezing point etc

2.4 State function or state variable : Variables like P, V, T are State Functions or State Variables because their values depend only on the present state of a system and not on how the state was reached

Condition for a function to be a state function :

(i) If ϕ is state function, $\int_A^B d\phi = \phi_B - \phi_A$

(ii) If ϕ is a state function, $\oint d\phi = 0$

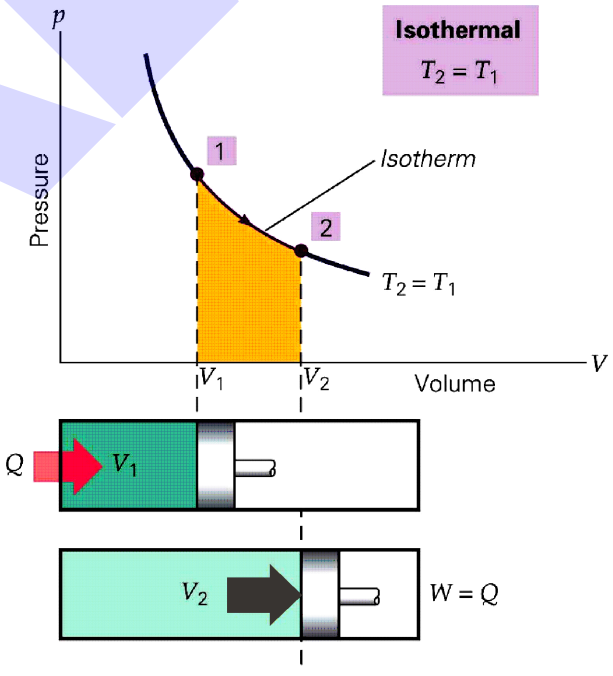
2.5 Path function : Function which depends on the path, i.e. how the process is carried out e.g. work & heat.

2.6 Thermodynamic process : A thermodymamic process involves change of a system from one state to another state.

Type of process :

(i) Isothermal process :

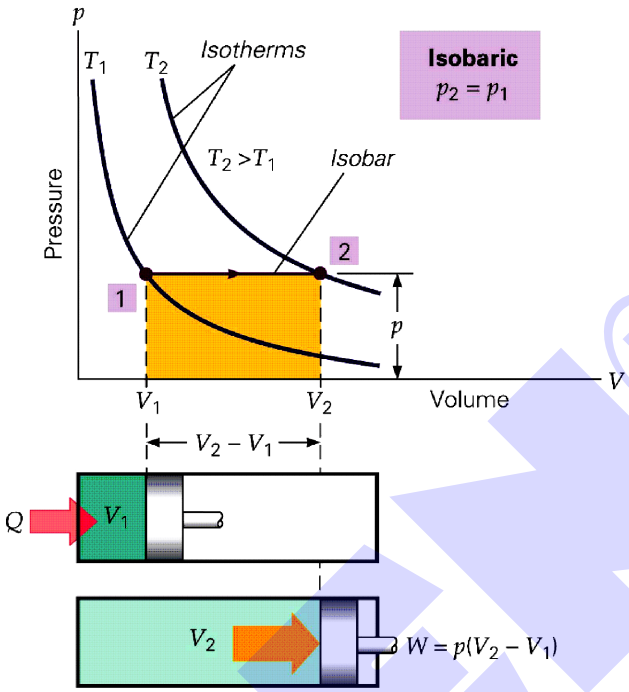
A process in which temperature of the system remains constant is called isothermal process.



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(ii) Isobaric process :

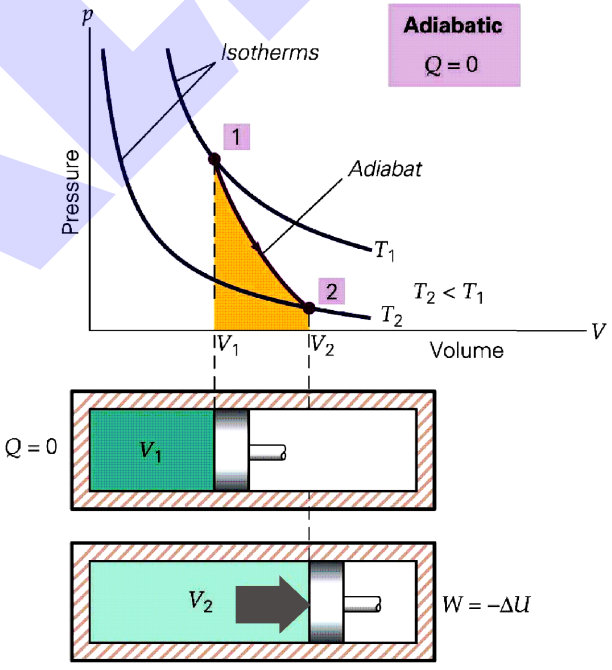
A process in which pressure of the system remains constant is called isobaric process. Temperature and volume of the system may change.



Ex. All the reactions or processes taking place in open vessel like boiling of water in open vessel, burning of charcoal, melting of wax take place at constant pressure (1 atm.)

(iii) Adiabatic process :

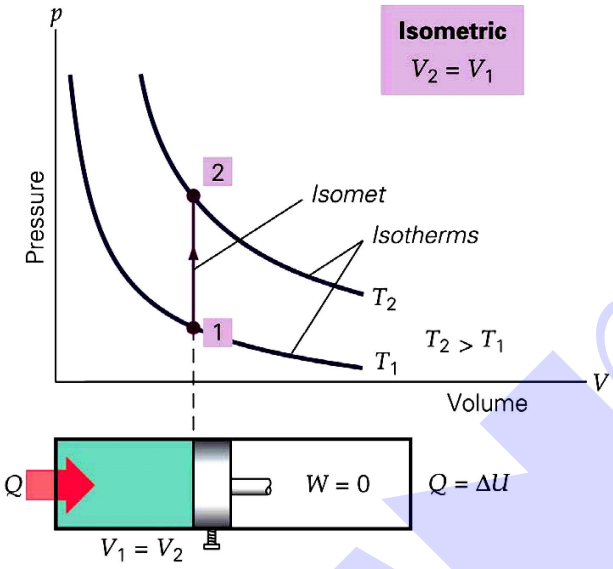
A process in which no heat exchange takes place is called adiabatic process. Adiabatic process occurs in systems with insulated walls.



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(iv) Isochoric process :

The process for which volume of the system remains constant is called isochoric process i.e., Heating of gas in closed or rigid vessel.



(v) Cyclic process : It is combination of two or more process in which the final state of system, becomes identical to the initial. The net change in all thermodynamic properties of system must be zero.

$\Delta T = \Delta P = \Delta V = \Delta H = \Delta S = \Delta G$ ..... $= 0$

However Q_{net} or W_{net} may or may not be zero.

(vi) Reversible or Irreversible process : If the initial state may be received just by reversing the direction of process at any state, process is called reversible. In irreversible process the initial state can never be achieved just by reversing the direction of process.

In the reversible process, the driving force is only infinitesimally greater than the opposing force. In the irreversible process, they differ largely.

A perfectly reversible process is a theoretical process because it takes infinite time but reversible processes are important because it results maximum efficiency in any machine. The actual process occurring in the machine is quasi static which may tend to reversible or irreversible process.

During irreversible process, one of the equation of state like PV = nRT, is valid. Such equation is valid only at initial & final state. However during reversible process such equation are valid throughout the process.

A process is reversible when the system throughout remain in thermodynamic equilibrium with the surrounding.

A system is said in thermodynamic equilibrium when it simultaneously satisfy the following equilibria.

(a) Thermal equilibrium: Same temperature throughout (No heat transfer within the system)

(b) Mechanical equilibrium : Same pressure or force throughout the system (No work should be performed by one part on the other)

(c) Material equilibrium : No change in the composition of system with time. No mass transfer within the system.

If a system is in thermodynamic equilibrium there is no net energy or mass transfer within the system.

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3. WORK

Thermodynamically, work may be defined as the form of energy which appears only when, there is some change in the boundary of the system. Such work is called mechanical or PV work.

Work may also be non-mechanical like electric work.

Presently, we will discuss only mechanical work.

(i) It is not thermodynamic property of system.

(ii) It depends on the quantity of system.

(iii) Sign convention, work on system = (+)ve.

In physics, work is calculate from the force applied by system, but in chemistry due to external force.

$$W = \int_{V_1}^{V_2} P \cdot dV = -\int_{V_1}^{V_2} P_{ext} dV$$

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4. HEAT

Heat is defined as the energy that flows into or out of a system because of a difference in temperature between the system and its surrounding.

According to IUPAC convention

heat lost by system is expressed with –ive sign

heat given to system is expressed with +ive sign

* $q_v = nC_{v,m}dT$
* $q_p = nC_{p,m}dT$
* $C_{p,m} - C_{v,m} = R$ (for ideal gas)
* C_v & C_p depends on temperature even for an ideal gas. ($C = a + bT + cT^2$.....)

General values of C_V & C_P for an ideal gas can be taken as follows.

Atomicity		n_{tr}	n_{Rot}	n_{Vib}	C_V		C_P		γ	
						Incl. Vib	Excl. Vib	Incl. Vib	Excl. Vib	Incl. Vib
Mono		3	0	0	$\frac{3}{2}R$	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$	$\frac{5}{3}$
Di		3	2	1	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{7}{5}$	$\frac{9}{7}$
Tri	Linear	3	2	4	$\frac{5}{2}R$	$\frac{13}{2}R$	$\frac{7}{2}R$	$\frac{15}{2}R$	$\frac{7}{5}$	$\frac{15}{13}$
	Non Linear	3	3	3	3R	6R	4R	7R	$\frac{4}{3}$	$\frac{7}{6}$

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5. INTERNAL ENERGY (E & U)

Every system having some quantity of matter is associated with a definite amount of energy, called internal energy.

$U = U_{Kinetics} + U_{Potential} + U_{Electronic} + U_{nuclear} + ...$

U is a state function & is an extensive property.

$\Delta U = U_{final} - U_{initial}$

For a given closed system

$U = f(T, V)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_v dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

6. ENTHALPY

Chemical reactions are generally carried out at constant pressure (atmospheric pressure) so it has been found useful to define a new state function **Enthalpy** (H) as.

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta(PV)$$

at constant pressure

$$\Delta H = \Delta U + P\,\Delta V$$

combining with first law.

$$\Delta H = q_p$$

Hence transfer of heat at constant volume brings about a change in the internal energy of the system whereas that at constant pressure brings about a change in the enthalpy of the system.

6.1 RELATIONSHIP BETWEEN ΔH & ΔU :

The difference between ΔH & ΔU becomes significant only when gases are involved (insignificant in solids and liquids)

$$\Delta H = \Delta U + \Delta(PV)$$

If substance is not undergoing chemical reaction or phase change.

$$\Delta H = \Delta U + nR\Delta T$$

In case of chemical reaction

$$\Delta H = \Delta U + (\Delta n_g)RT$$

7. ZEROTH LAW OF THERMODYNAMICS

Two system in thermal equilibrium with a third system are also in thermal equilibrium with each other. It introduces temperature as a state function.

8. FIRST LAW OF THERMODYNAMICS

"Total energy of universe remains constant." It is law of conservation of energy.

Let us consider a system whose internal energy is U_1. If the system is supplied with heat q, the internal energy of the system increases to $U_1 + q$. If work (w) is done on the system, the internal energy in the final state of the system, U_2 is given by

$$U_2 = U_1 + q + w$$

or $$U_2 - U_1 = q + w$$

$$\Delta U = q + w$$

According to IUPAC, heat added to the system and work done on the system are assigned positive values as both these modes increase the internal energy of the system.

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9. CALCULATION OF q, w, ΔU & ΔH IN VARIOUS PROCESS :

9.1 ISOTHERMAL EXPANSION OR COMPRESSION OF IDEAL GAS :

For isothermal process : $\Delta T = 0$

$\therefore$ $\Delta U = n.C_{p,m}.\Delta T = 0$

$\Delta H = n.C_{p,m}.\Delta T = 0$

and $q = \Delta U - w = -w$

Now, w depends on path (reversible or irreversible).

9.1.1 Reversible process :

$$w_{rev.} = -\int_{V_1}^{V_2} P_{ext} \cdot dV = -\int_{V_1}^{V_2} (P \pm dP) \cdot dV = -\int_{V_1}^{V_2} P \cdot dV = -\int_{V_1}^{V_2} \frac{nRT}{V} \cdot dV$$

$$\therefore \quad w_{rev.} = -nRT \cdot \ln \frac{V_2}{V_1} = -nRT \cdot \ln \frac{P_1}{P_2}$$

9.1.2 Irreversible process, against a constant external pressure :

$$w_{irr.} = -P_{ext} \int_{V_1}^{V_2} dV = -P_{ext}(V_2 - V_1)$$

9.1.3 Free expansion (or expansion in vacuum) :

$P_{ext} = 0$ but dV = finite, Hence, w = 0

9.1.4 Comparison of magnitude of work :

In expansion, magnitude of work $= -w_{exp}$

In compression, magnitude of work $= w_{comp}$

Now, $w_{rev} - w_{irr} = (-P.dV) - (-P_{ext}.dV) = (P_{ext} - P).dV$

$= -$ve, always

In expansion, $P_{ext} < P$ and dV = +ve

In compression, $P_{ext} > P$ and dV = –ve

Hence, $w_{rev} < w_{irr}$ (Always, with proper sign).

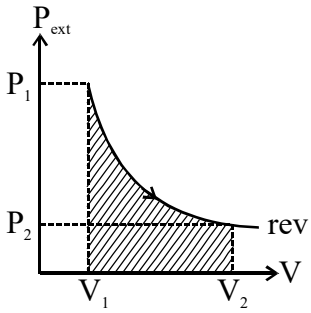
Now, as magnitude of work in exapansion is $-w_{exp}$, hence, $(-w_{rev, exp}) > (-v_{irr, exp})$, i.e., magnitude of work is greater when expansion is reversible.

And, as magnitude of work in compression is w_{comp}, hence, $(w_{rev, comp}) < (w_{irr, comp})$, i.e., magnitude of work is greater when compression is irreversible.

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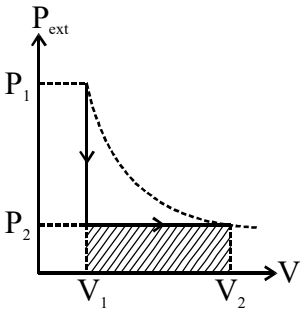
9.1.5 Graphical comparison :

(a) Expansion :

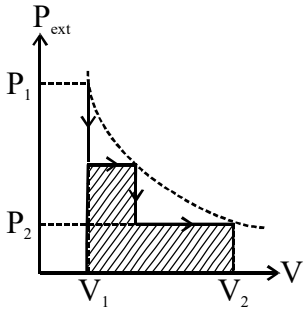


$(P \approx P_{ext})$

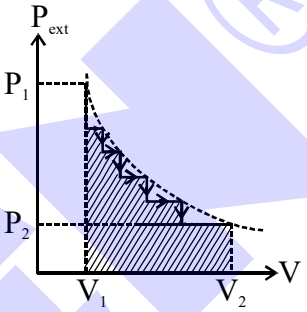
(Reversible)



(Single step irreversible)



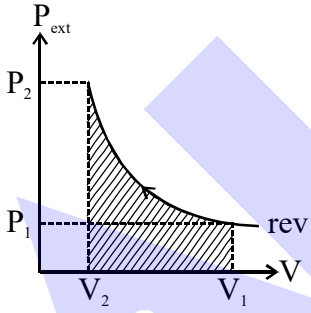
(Two step irreversible)



(n step irreversible)

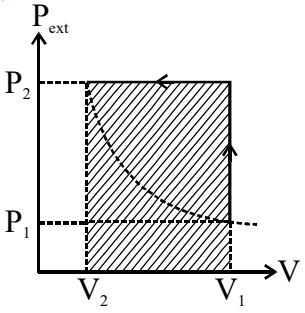
$(-w_{exp})$: Rev > n step > ..... > 2 step > single step

(b) Compression :

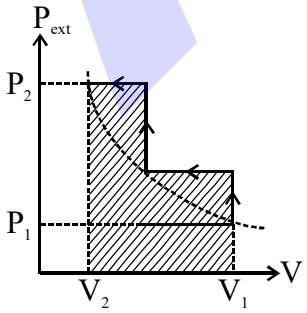


$(P \approx P_{ext})$

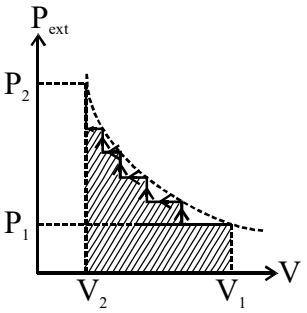
(Reversible)



(Single step irreversible)



(Two step irreversible)



(n step irreversible)

(w_{comp}) : Rev < n step < ..... < 2 step < single step

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Ex.1 Two moles of an ideal gas undergoes isothermal expansion from 4L to 20L at 27°C. Calculate q, w, ΔU and ΔH, if the process is performed.

(i) reversibly

(ii) irreversibly, against a constant external pressure of 1atm.

(iii) as free expansion

Sol. For isothermal process, $\Delta T = 0$. Hence,

$\Delta U = n.C_{v,m}.\Delta T = 0$

$\Delta H = n.C_{p,m}.\Delta T = 0$

(i) $w = -nRT \cdot \ln \frac{V_2}{V_1} = -2 \times 8.314 \times 300 \times \ln \frac{20}{4} = -8028.52J$

and $q = -w = 8028.52$ J

(ii) $w = -P_{ext}.(V_2 - V_1) = -1 \text{ atm}(20L - 4L) = -16\text{L-atm}$

$= -16 \times 101.3J = -1620.8J$

$q = -w = 1620.8J$

(iii) $w = 0$ and $q = 0$

9.2 ADIABATIC EXPANSION OR COMPRESSION OF IDEAL GAS :

$q = 0$

$\therefore \quad \Delta U = w = n.C_{v,m}.(T_2 - T_1)$

and $\Delta H = n.C_{p,m}.(T_2 - T_1)$

The change in temperature may be calculated as

$$n \cdot \int_{T_1}^{T_2} C_{v,m} \cdot dT = -\int_{V_1}^{V_2} P_{ext} \cdot dV$$

(a) If $C_{v,m}$ is temperature independent and the process is reversible, then

$T.V^{\gamma - 1} = \text{constant}$

or, $P.V^{\gamma} = \text{constant}$

or, $T^{\gamma} P^{1-\gamma} = \text{constant}$

(b) If $C_{v,m}$ is temperature independent and the process is irreversible, then

$n.C_{v,m}.(T_2 - T_1) = -P_{ext}.(V_2 - V_1)$

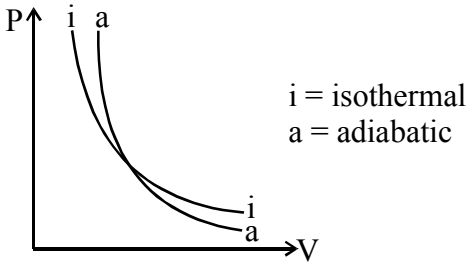
(i) The temperature of ideal gas decreases in adiabatic expansion (except free expansion, which is isothermal too) and the temperature of ideal gas increases in adiabatic compression.

(ii) The final temperature of gas is always higher in irreversible process (expansion or compression) relative to reversible process, for the same change in volume.

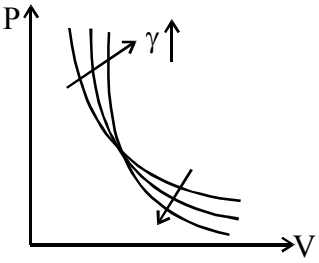
(iii) Just like isothermal process, the magnitude of work is maximum when expansion is reversible and compression is irreversible (single step).

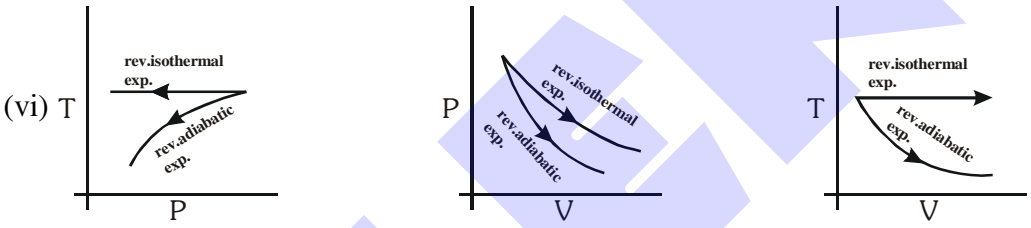
(iv) Work in reversible isothermal and adiabatic processes may be compared graphically.

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(v) Work in reversible adiabatic process may be compared graphically.





Comparison of reversible adiabatic expansion and reversible isothermal expansion on TP, PV and TV diagram

Ex.2. Two moles of an ideal monoatomic gas undergoes adiabatic expansion from 5L, 127°C to 40L. Calculate q, ΔU, w and ΔH, if the process is performed.

(i) reversibly

(ii) irreversibly, against a constant external pressure of 0.1atm.

(iii) as free expansion

Sol. : For adiabatic process, q = 0

(i) $T \cdot V^{\gamma-1} = \text{constant} \Rightarrow T_1 \cdot V_1^{\gamma-1} = T_2 \cdot V_2^{\gamma-1}$

$$\therefore \quad T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 400 \times \left(\frac{5}{40}\right)^{\frac{5}{3}-1} = 100\text{K}$$

Now, $w = \Delta U = nC_v\Delta T = -7482.6$ J

$\Delta H = \gamma\Delta U = -12471$ J

(ii) $\Delta U = w = -P_{ext}.(V_2 - V_1) = -0.1 \times (40 - 5) \times 101.3 = -354.55$J

and $\Delta H = \frac{5}{3} \cdot \Delta U = -590.92$J

(iii) $\Delta U = w = 0$

and $\Delta H = 0$

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Ex.3. Five moles of an ideal monoatomic gas undergoes adiabatic expansion from 12 atm to 1 atm, against a constant external pressure of 1 atm. If the initial temperature of gas is 27°C, calculate the final temperature. Also calculate q, ΔU, w and ΔH.

Sol. : $q = 0$

$\Delta U = w$

or, $n.C_{v,m}.(T_2 - T_1) = -P_{ext}.(V_2 - V_1)$

or, $n.\frac{3}{2}R(T_2 - T_1) = -P_2\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$

or, $\frac{3}{2}(T_2 - 300) = -\left(T_2 - 300 \times \frac{1}{12}\right) \Rightarrow T_2 = 190K$

or, $\Delta U = w = 5 \times \frac{3}{2}R \times (190 - 300) = -6859.05J$

and $\Delta H = \gamma.\Delta U = -11431.75J$

9.3 ISOBARIC PROCESS FOR AN IDEAL GAS :

$q = \Delta H = n.C_{v,m}.\Delta T$

$\therefore \quad \Delta U = n.C_{v,m}.\Delta T$

$w = -P.(V_2 - V_1) = -nR.\Delta T$

Ex.4. One mole of an ideal diatomic undergoes isobaric expansion fromo 27°C to 87°C. Calculate q, ΔU, w and ΔH.

Sol. : $q = \Delta H = n.C_{v,m}.\Delta T = 1 \times \frac{7}{2}R \times 60 = 1745.94\ J$

$\Delta U = \frac{\Delta H}{\gamma} = 1247.1J$

and $w = -nR.\Delta T = -1 \times 8.314 \times 60 = -498.84J$

9.4 ISOCHORIC PROCESS FOR AN IDEAL GAS :

$w = 0$

$q = \Delta U = n.C_{v,m}.\Delta T$

$\therefore \quad \Delta U = n.C_{v,m}.\Delta T$

Ex.5. Ten moles of an ideal gas ($\gamma = 1.2$) is heated from 27°C to 47°C at constant volume. Calculate q, ΔU, w and ΔH.

Sol. : $w = 0$

$q = \Delta U = n.C_{v,m}.\Delta T = 2 \times \frac{R}{1.2 - 1} \times 20 = 200\ R$

$\Delta H = \gamma.\Delta U = 240R$

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9.5 POLYTROPIC PROCESS FOR AN IDEAL GAS :

$\Delta U = n.C_{v,m}.\Delta T$

$\therefore \quad \Delta H = n.C_{p,m}.\Delta T$

For ideal gas is reversible polytropic process and $C_{v,m}$ temperature independent,

$P.V^x = \text{constant}$

or, $T.V^{x-1} = \text{constant}$

where x = polytropic index

$$w_{rev} = \frac{P_2V_2 - P_1V_1}{1-x} = -\frac{nR(T_2 - T_1)}{1-x} \qquad (x \neq 1)$$

$$= -nRT \cdot \ln\frac{V_2}{V_1} \qquad (x = 1)$$

and $q = n.C_m.\Delta T$

The molar heat capacity, $C_m = C_{v,m} + \frac{P \cdot dV}{n \cdot dT} = C_{v,m} + \frac{R}{1-x}$

9.6 CYCLIC PROCESS :



9.7 CHANGE IN PHYSICAL SATE :

Solid $\rightleftharpoons$ liquid

Liquid $\rightleftharpoons$ Gas

Solid $\rightleftharpoons$ Gas

Changes in physical state occurs at constant pressure and temperature conditions.

$\Delta U = q + w$

$q = \Delta H = m.L$

and $w = -P(V_{\text{final state}} - V_{\text{initial state}})$

Ex.6. 90 gm water is completely converted into steam at 100°C and 1 atm. Calculate q, ΔU, w and ΔH. Latent heat of vaporisation of water at 100°C is 540 cal/gm.

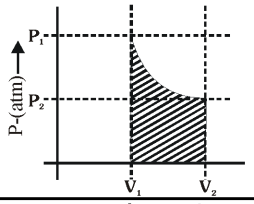
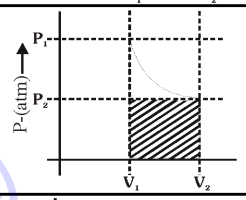
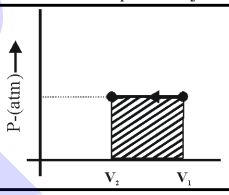
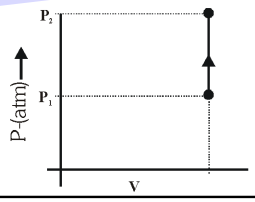
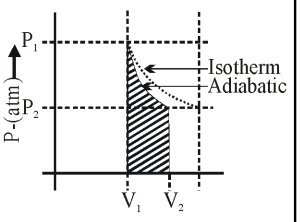
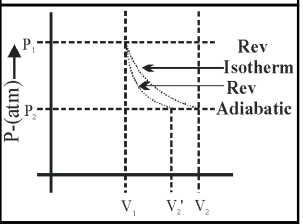
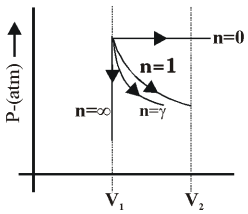
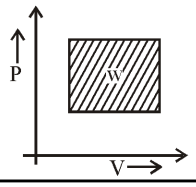
Sol. : $\Delta H = m.L = 90 \times 540 = 48600$ cal

$$w = -P(V_{vap} - V_{water}) = -P.V_{vap} = -nRT = -\frac{90}{18} \times 2 \times 373 = -3730 \text{ cal}$$

$\Delta U = q + w = 48600 + (-3430) = 44870$cal

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10. SUMMARY

Process	Expression for w	Expression for q	ΔU	ΔH	Work on PV-graph
Reversible isothermal	$w = -nRT\ln\frac{V_2}{V_1}$ $= -nRT\ln\frac{P_1}{P_2}$	$q = nRT\ln\left(\frac{V_2}{V_1}\right)$ $q = nRT\ln\left(\frac{P_1}{P_2}\right)$	0 process	0	
Irreversible isothermal	$w = -P_{ext}(V_2 - V_1)$ $= -P_{ext}\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$	$q = P_{ext}(V_2 - V_1)$	0	0	
Isobaric process	$w = -P_{ext}(V_2 - V_1)$ $= -nR\Delta T$	$q = \Delta H = nC_P\Delta T$	$\Delta U = nC_V\Delta T$	$\Delta H = nC_P\Delta T$	
Isochoric process	$w = 0$	$q = \Delta U = nC_V\Delta T$	$\Delta U = nC_V\Delta T$	$\Delta H = nC_P\Delta T$	
Reversible adiabatic process	$w = nC_V(T_2 - T_1)$ $= \frac{P_2V_2 - P_1V_1}{\gamma - 1}$	$q = 0$ PV^{γ}=constant $TV^{\gamma-1}$=constant $TP^{1-\gamma/\gamma}$=constant	$\Delta U = nC_V\Delta T$	$\Delta H = nC_P\Delta T$	
Irreversible adiabatic process	$w = nC_V(T_2 - T_1)$ $= \frac{P_2V_2 - P_1V_1}{\gamma - 1}$	$q = 0$ $nC_v(T_2 - T_1) =$ $-P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$	$\Delta U = nC_V\Delta T$	$\Delta H = nC_P\Delta T$	
Polytropic	$w = \frac{P_2V_2 - P_1V_1}{n - 1}$ $w = \frac{R(T_2 - T_1)}{(n-1)}$	$q = \int_{T_1}^{T_2} C_V dT$ $w = \frac{R(T_2 - T_1)}{(n-1)}$	$\Delta U = nC_V\Delta T$	$\Delta H = nC_P\Delta T$	
Cyclic Process	Area encolsed in PV-diagram For clockwise –ive For anticlockwise +ive	$q = -w$	0	0	

11. LIMITATION OF FIRST LAW :

It can predict the change in energy as a result of change in state or vice-versa, but can not predict the natural direction of change (whether a change can happen on it's own or not).

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MISCELLANEOUS PREVIOUS YEARS QUESTION

1. A sample of argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 dm^3 to 2.50 dm^3. Calculate the enthalpy change in this process. $C_{v,m}$ for argon is 12.48 JK^{-1} mol^{-1} **[JEE 2000]**

Ans. **$\Delta H \simeq -115$ J**

$T_1V_1^{\gamma-1} - T_2V_2^{\gamma-1}$

$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$T_2 = 300\ (1/2)^{8.3/12.48} \quad \left[\gamma^{-1}\frac{R}{C_{\gamma.m}}\right]$$

$T_2 = 189.2$ Kelvin.

$$n = \frac{P_1V_1}{RT_1} = \frac{(1)(1.25)}{(0.0821)(300)} = 0.507$$

$\Delta H = nC_{p.m}\ (T_2 - T_1)$

$= 0.0507\ (20.78)\ (189.2 - 300)$

$= -116.732$ Joule.

2. Which of the following statement is false ? **[JEE 2001]**

(A) Work is a state function

(B) Temperature is a state function

(C) Change of state is completely defined when initial and final states are specified.

(D) Work appears at the boundary of the system

Ans. **(A)**

3. One mole of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy (ΔU) = 30.0 L-atm. The change in enthalpy (ΔH) of the process in L-atm. **[JEE 2002]**

(A) 40.0

(B) 42.3

(C) 44.0

(D) Not defined, because pressure is not constant

Ans. **(C)**

$\Delta H = \Delta U + P_2Y_2 - P_1Y_1$

$\Delta H = 30 + 20 - 6$

$= 44$

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4. One mole of a liquid (1 bar, 100 ml) is taken in an adiabatic container and the pressure increases steeply to 100 bar. Then at a constant pressure of 100 bar, volume decrease by 1 ml. Find ΔU and ΔH.

[JEE 2004]

Ans. $\Delta U = 10$ J , $\Delta H = 990$ J

$0 + w = \Delta U$

$\Delta U = -P_{ext}\,(Y_2 - V_1)$

$= -100\left(\frac{-1}{1000}\right)$

$= 0.1$ bar litre.

$\Delta H = \Delta U + P_2Y_2 - P_1Y_1$

$\Delta H = 0.1 + \left[\frac{100(99) - (1)(100)}{1000}\right]$

$\Delta H = 0.1 + 9.8$

$\Delta H = 9.9$ bar. litre

$\Delta H = 990$ Joules

5. Two mole of an ideal gas is expanded isothermally and reversibly from 1 litre to 10 litre at 300 K. The enthalpy change (in KJ) for the process is :- **[JEE 2004]**

(A) 11.4 kJ (B) – 11.4 kJ (C) 0 kJ (D) 4.8 kJ

Ans. (C)

In isothermal process $\Delta T = 0$

$\Delta H = nC_{v.m}\Delta T$

$\Delta H = 0$

6. One mole of monoatomic ideal gas expands adiabatically at initial temp. T against a constant external pressure of 1 atm from one litre to two litre. Find out the final temperature.

($R = 0.0821$ litre. atm K^{-1} mol^{-1}) **[JEE 2005]**

(A) T (B) $\frac{T}{(2)^{\frac{5}{3}-1}}$ (C) $T - \frac{2}{3\times0.0821}$ (D) $T + \frac{2}{3\times0.0821}$

Ans. (C)

$-P_{ext}\,(V_2 - V_1) = nC_{v.m.}(T_2 - T_1)$

$-1(2 - 1) = 1\left(\frac{3}{2}R\right)(T_2 - T)$

$T_2 = T - \frac{2}{3R}$

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7. The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one.

(A) $\frac{4}{2}R$ (B) $\frac{3}{2}R$ (C) $\frac{5}{2}R$ (D) zero **[JEE 2006]**

Ans. (A)

$$\frac{P}{V} = C$$

$$PV^{-1} = C$$

$$C_m = C_{v.m.} + \frac{R}{1-(-1)}$$

$$= \frac{3}{2}R + \frac{R}{2} = \frac{4R}{2}$$

8. Among the following, the state function(s) is (are) **[JEE 2009]**

(A) Internal energy

(B) Irreversible expansion work

(C) Reversible expansion work

(D) Molar enthalpy

Ans. (A,D)

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